

Shahjalal University of Science and Technology

Department of Computer Science and Engineering



EMI MANAGEMENT AND LOAN TRACKING SYSTEM FOR SMALL BUSINESSES AND MICRO-LOAN PROVIDERS

A Project submitted to the Department of Computer Science and Engineering, Shahjalal University of Science and Technology, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

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is under my supervision. I, hereby, agree that the project can be submitted for examination.

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Certificate of Acceptance of the Thesis/Project

The thesis/project entitled *EMI Management and Loan Tracking System for small business and micro loan providers* submitted by the students

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Abstract

In Bangladesh, the micro-credit and small retail business sectors are dependent on manual, paper-based systems of dealing with Equated Monthly Instalment (EMI) and loan-tracking. Such a conventional method has a high probability of human error, embezzlement and lack of transparency which usually creates conflicts and loss of revenue. The project outlines the design and development of the full-fledged EMI Management and Loan Tracking System that is specially designed to meet the needs of the small businesses and the micro loan providers. The proposed system will fill the gap between the offline shop operations and the online digital trade by providing a single platform. The most important are the independent registration of sellers, automatic creation of the EMI schedule, a high-quality KYC verification system with the NID/Passport upload, and a risk analysis system that allows separating the borrowers into categories by their behavioral patterns under the payment agreements. Technically, the system is constructed based on three tier architecture of React.js front end, Node.js back end and MySQL database to ensure ACID compliance of financial transactions. Local peculiarities, like automatic SMS notifications, the ability to use a hybrid (online/offline) approach to creating loans, and inbuilt payment gateways (bKash, Nagad) prevent the need to deal with local issues. The system will benefit by digitizing these financial processes to promote financial inclusion, minimizing operational risks to small sellers, and a clear, trust-based ecosystem of lending.

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Chapter 1

Introduction

Small scale sellers and a large number of small businesses offer products on the basis of loan based EMI in Bangladesh. Customers make down payment and settle the balance in instalments. Nonetheless, these sellers largely depend on paper notebooks, basic spreadsheet, or memory-based tracking that lead to the frequent error, payment lapses, disputes, and mismanagement of finances at large.

In this project, the researcher intends to overcome these obstacles by formulating an all-inclusive EMI and Loan Management System that will target the small and medium sellers. The system will computerize the whole work process including the addition of products, verification of buyers and loan giving alongside handling EMI payments, monitoring overdue payments and the production of reports. Sellers can also enter items in their online catalog where the buyer has a choice of purchasing them with full payment or even EMI and there will be the smooth e-commerce experience will be offered with the loan services offered.

The system is likely to substitute the manual processes with automation and transparency in selling and lending loans by agreeing with the real-life demands of the Bangladeshi market, rendering the EMI-based selling and lending processes more effective and trustworthy.

1.1 Contribution to the Society

The EMI and Loan Management System will have a great impact on the society in a number of ways:

1. **Empowering Small businesses:** The system assists the sellers in handling the loans and EMI records electronically, eliminating errors, fraud, and enhancing financial security.
2. **Financial Inclusion:** Numerous low-income consumers use loans or EMIs to get the basic commodities (e.g., phones, electronics, and furniture). The system renders the

loan management systematic and transparent which is in favor of both the sellers and buyers.

3. **Improvement of Trust and Transparency:** The presence of such options as the KYC verification, automatic EMI bills, receipts, and buyer notifications promotes trust and decreases conflicts.
4. **Reducing of Fraud by Identity Verification:** NID or passport images can be stored and checked safely by the sellers, which will help in avoiding cases of fake identities when issuing loans.
5. **Promoting Digital Transformation:** Lack of modern software is common among small sellers because it is usually expensive. This system is a cheap, user-friendly system that can propel them to online business management.

1.2 Expected Outcome

The system will go a long way to enhance the management of EMI-based sales and lending to the small businesses. Automated EMI calculation, flexible loan creation, digital payment tracking, overdue analysis, and online shopping will help to have a more convenient and transparent process by the sellers and buyers.

Customers will be allowed to view products online and decide on the option of making them purchases either with full payment or EMI (loan-based) options. The sellers will enjoy less paperwork, proper tracking, and financial planning.

In general, the objective of the project is to minimize human error, enhance the trust between the sellers and buyers, and provide an efficient and scalable loan and EMI management system to small businesses in Bangladesh.

1.3 Usability of the Product

The simplicity, accessibility and user friendliness are the priorities in the development of the system. The system is easy to use and sellers and buyers who have limited knowledge in technology can use the system with ease due to its user-friendly interface and simple navigation. The fact that it can be supported by both English and Bengali makes it wider.

The platform is compatible with mobile phones, tablets and computers, and thus it is applicable among shopkeepers, micro-loan providers, as well as buyers. AI-assisted reminders, cloud storage, automated workflows in loans and improved ease of use make the system even easier to use since users can comfortably handle their EMI and loan transactions.

Chapter 2

Background Study

2.1 Problem Statement

Bangladesh small businesses that facilitate the sale of goods and services on EMI or loan base have a lot of operational difficulties owing to unintegrated digital management system. Majority of the sellers rely on notebooks, paper receipts and simple spreadsheets as a means of keeping a track of customer loan, EMI schedule and payments. Such manual processes are time-consumable as well as subject to errors leading to loss of money and customer experience.

The core issues that provide a reason to need an all-out EMI and loan management solution include:

- **Frequent manual errors:** The sellers frequently compute the interests, create EMI schedules and use the late fees manually. This results in a high level of mistakes, inconstancy, and conflicts. A 15-20% business loss in terms of revenue per year which may have a direct impact on the sustainability of business could arise out of little mistakes repeated every single month among a large number of buyers.
- **Disconnected Offline and Online Business:** The majority of small business sell in their offline shops but they desire to go online. Nevertheless, they do not have a single system that coordinates the two channels.
- **Poor Collection and Follow-Up Management:** EMI collection was dependent on manual follow-up that was a waste of time (Phone calls or messages). Buyers can easily forget the due dates as they have no automatic reminders and notifications and thus make late payments. Sellers are also not able to determine who are always late or prone to default their payments.
- **Minimal Business Intelligence:** Small sellers have no analytics and insights. They are not able to get performance metrics, sales and collection trends, credit risk scores,

forecasting tools. This incomplete data does not allow sellers to know the development of their business, who are the high-risk consumers, and to optimize their strategies.

2.2 Objectives

The main aim of EMI and Loan Management System is that:

1. **Develop a Unified EMI Platform:** Develop one platform that will provide the ability to sell both in-store (offline) and online (marketplace) products and track EMI without having separate systems.
2. **Simplify EMI Management:**
 - (a) In the case of offline sales, EMI plans can be prepared manually by the sellers.
 - (b) In case of online sales, EMI schedules are created automatically as an order is approved by the seller.
3. **Flexible Payments:** Accept various payment methods such as cash payment by seller, online payments by use of bKash, Nagad or SSLCommerz, partial and advance payments.
4. **Smart Reminder System:** Automated SMS and in-app notifications to buyers- before it is due, on it is due, and when it is overdue.
5. **Business Insights & Reporting:** Gives useful insights like, overdue trends and risk scores, Collection performance, Reports which can be exported as PDF or Excel.

2.3 Existing Solution

There are various solutions in the market, including enterprise banking system (Finacle, Temenos), micro-finance software (Mifos, Musoni), and gear towards small business (QuickBooks, Zoho Books, Square). However:

- Enterprise solutions are expensive (USD 50,000+) and not easy to use and need specific IT personnel so they are not suitable to small business.
- Micro-finance platforms are institution-based with a high level of set-up cost (BDT 100,000+), without individual seller and peer-peer marketplaces.
- Accounting and POS will receive general invoicing, but will not provide EMI specific options like installment calculator, payment notices, and KYC verification.

There is no solution that will manage the EMI, the customer approval and the trusted marketplace in which small retailers will operate.

2.4 Target User and Stakeholders

Primary Users: At Bangladesh, small-business sellers, shop owners, and NGOs, as well as individual purchasers who buy goods via EMI or offer installment-based sales upkeep, are involved. **Stakeholders:**

1. The owners of small businesses and retailers who use the system to control offline and online EMI sales, installments and minimize manual errors.
2. Buyers/customers, customers who buy products (through EMIs and otherwise) online (or offline) and wish to know the payment schedule (clear and transparent) with reminders.
3. System administrators, that manage the work of platforms, guarantee the protection of data and assist the onboarding of multi-sellers.
4. Partners in the payment gateway (bKash, Nagad, SSLCommerz) that facilitate safe transactions online by buyers.
5. Micro-loan NGOs and community sellers do require appropriate KYC documentation, and legal history of installment product sales.

2.5 Scope and Limitations

Scope: The main features are buyer KYC (NID/Passport upload) verification, EMI plan creation, online/offline payment, overdue tracking, automated reminders, seller products listing, and buyer portal to view EMIs and receipts.

Limitations:

- KYC verification performed manually can be delayed and not done correctly.
- Online facilities rely on reliable internet and third party payment gateways.
- No credit check services or automated government ID.

2.6 Gap Analysis

Identified Gaps

Gap	Existing Solutions	Our Solution
Affordability	BDT 10,000-50,000/year	Low-cost/Free tier
Independent Registration	Admin-only account creation	Self-registration for all
Hybrid Management	Separate systems for offline/online	Unified platform
Marketplace	No integrated product listing	Built-in marketplace
Localization	International focus	Bangladesh payment gateways
Ease of Use	Complex interfaces	Simple, intuitive design
Offline Support	Online-only	Manual entry for shop sales

Table 2.1: Solution Comparison

2.7 Technology Overview

The system will use:

1. React.JS: To create a responsive, fast seller portal and buyer portal.
2. Node.js: To serve backend logic, EMI computations, authentication, and APIs.
3. MySQL Database: To store users, loans, EMIs, payments and KYC records safely.
4. Cloud Storage (e.g., Google Drive/S3): To store uploaded NID/Passport images in a safe place.
5. Payment Gateway APIs: bKash or Stripe or Nagad or SSLCommerz online payment.
6. Firebase Cloud Messaging (FCM): To provide automatic due notifications and reminders.

Chapter 3

Related Works

Digital transformation has seen the field of small business financial management and credit tracking change tremendously. A number of systems and platforms that are already in place have resulted in EMI management, installment tracking and micro-finance solutions. Nevertheless, most of the solutions are for big companies or general accounting requirements, and no one caters to the small retailers who require specific EMI and marketplace solutions. Some of the eminent works and platforms regarding the proposed project are given below:

3.1 Existing Systems

3.1.1 Finacle (Infosys)

- **Features:** Complete banking core system, Loan management, EMI-calculations, multi-currency and strong security measures.
- **Weaknesses:** Very expensive (USD 50,000- and above), requires special IT infrastructure, multi-layered, not designed for small retailers.

3.1.2 Mifos X

- **Features:** Open-source micro-finance system Lending tracking, repayment schedule Customer management Support branch operations
- **Weaknesses:** High set up fees (BDT 100,000+), no marketplace, technical knowledge, and are not applicable to individual sellers.

3.1.3 QuickBooks

- **Features:** Invoicing, Expense Tracking, Financial Reporting, Payment Reminders, Multi User.

- **Weakness:** Generic accounting focus, no EMI specific calculation, not KYC verified, no customer credit scoring, no market place features.

3.1.4 Zoho Books

- **Features:** Accounting online, create an invoice, automation, payment process, mobile application.
- **Weaknesses:** Is not installment tracking, does not have any EMI management, does not have the tools to verify its customers, charged by subscription.

3.1.5 Square POS

- **Features:** easy to use interface, payment processing, inventory, sales analysis, customer profiles.
- **Weakness:** Done once transactions, no EMI calculation, no installment tracking and payment reminders, transaction fees are charged.

Chapter 4

Requirement Analysis

4.1 Functional Requirements

FR1: Independent Seller Registration

- Registration of personal data (name, owner, email, phone, address, type).
- Email verification via OTP which expires in 10 minutes
- Account which was created with the pending status that is pending approval from the admin.
- Password validation: minimum 8 characters with uppercase, number, special character Unique email and phone numbers enforcement.

FR2: Independent buyer registration

- Simplified registration with name, email/phone, and password
- Email/SMS verification for instant account activation
- No admin approval required for immediate marketplace access
- Full profile completion (address, NID) required only when requesting EMI

FR3: Admin approval for sellers

- Dashboard of the administrator to check on the pending registration of the seller.
- Actions: Approve, Reject (reason) or Seek further information.
- Automatic email notification of every action of approving.
- Complete accountability audit trail recording

FR4: Upload and checking of document

- Upload customer identity documents (NID front/back or Passport) with optional selfie.
- Accepted Formats: JPG, PNG, PDF
- Manual review of seller, approve/ reject workflow.
- Secure files with random files outside web root.
- Refusal brings opportunity of re-uploading with cause. All access to documents should have an audit trail.

FR5: Product Management

- CRUD products (name, description, category, price, stock, up to 5 images) Automatic stock tracking as well as decrement by sales.
- Low stock alerts for sellers
- Search and filter (category, price range, seller) Buyer market place. Description, pictures, details about the product, information of the seller.

FR6: Creating EMI Offline

- Choose verified customer and product or type custom amount.
- Enter down payment, interest rate, tenure (3-60 months), interest type.
- System computes EMI and generates entire amortization schedule. SMS/ email payment schedule in the hands of customer.
- Manual confirmation by seller before activation

FR7: EMI Auto-Generation for On-line.

- Buyer decides to make purchases with or without EMI at market place.
- Makes wishes in terms of EMI preview down payment, tenure etc.
- Buyer creates a profile and Seller received notification
- On approval: Emi autogenerated, schedule generated, down payment link sent. Considering the confirmation of down payment, EMI is automatically activated.

FR8: Calculation of Interests in a flexible manner

- Three ways of calculating were supported:
- Flat rate: Interest on original principal throughout the period

- Reducing Balance: Interests on outstanding principal.
- $EMI = \frac{P \times r \times (1+r)^n}{(1+r)^n - 1}$
- Zero interests: There is no interest in promotional offers.
- Breakdown of installments, principal and interest.
- Adjustable late charge policies - Fixed Amount, daily charge, or EMI percentage.
Computerized application of late fee and overdue installment.

FR9: Recording of cash manually

- Enter off line payment amount, date, mode (Cash/Bank Transfer/Cheque). Partial and advance payment and installment support.
- Automatically produces receipt in PDF format with QR code.
- Print/Email/SMS receipt.

FR10: Gateway Integration

- Integration to bkaash, Nagad, Stripe and SSLCommerz.
- Automated installment status credentials and receiving.

FR11: Auto Reminders

- Checks payment schedule every day at 9:00AM.
- Pre-Due Reminders - 3 days before till SMS + Email with payment link.
- Due-Date Reminders: Due date SMS (In-app notification) + Email + In-app notification
Overdue notification 1, 3 days, 7 days, 15 days, 30 days after through SMS + Email + In-app (red badge)
- Preferences (enable/disable channels, time, language), which can be set by the user.
Time and again SMS delivery tracking.

FR12: Overdue Analysis: System sort the overdue accounts into buckets of risk:

- Low Risk: 0-5 days overdue
- Medium Risk: 6-15 days overdue
- High Risk: 16-30 days overdue
- Critical: 30+ days overdue

- Calculation of risk score : $(\text{Overdue Amount} * \text{number of total days overdue}) / (\text{Total EMIs} * \text{average EMI})$.

FR13: Analytics and Exports

- Summary cards Dashboard (active EMIs, due amount, number of overdues, monthly collections)
- Charts: collections trend, risk categories that overdues are, sales by products.
- Custom reports: Collection, Overdue, Sales, EMI Summary.
- Export formats are PDF (formatted) and Excel (data).
- Filters: product, status, date range, and customer.

FR14 Automation of the backup system

- The incremental backups will be done daily, at the 2:00AM and the full backups will be done weekly on Sundays.
- Data contained in the backups, database, KYC documents, product pictures, receipts, config files.
- Integration of google drive API 90 days retention.

4.2 Non Functional Requirements

Performance Requirements:

- With normal internet connections, the application will take 3 seconds to load the dashboard.
- The system shall support up to 5,000 users at a time without affecting its performance.
- Results of queries on databases involving EMI calculations and payment schedules will be availed within a time span of 2 seconds.
- The generation of reports (PDF/Excel) will take less than 10 seconds and up to 1,000 records.

Usability:

- The user interface will offer an easy navigation to access any feature and limit the number of clicks to the maximum of 3.
- Mobile-friendly design between 320 to 1920 screen.

- Tips on help that are context relevant and straightforward error messages and recommendations.

Scalability:

- The application code will be designed to add new functionality, like support new APIs or new wearables, with minimum code modifications.
- The increase in the number of users by 300 percent will be supported by the backend system without much effect on the system performance.
- Database will support 100, 000 EMI records optimally indexed.

Security:

- Sensitive data stored and transmitted via AES-256 and TLS 1.2+ respectively.
- JWT-based authentication and expiration of a token and lockout of an account after 5 attempted logins.
- KYC files saved with randomized filenames are located outside of web root.
- Audit records to be stored during 2 years in accordance with local data protection laws.

Availability:

- The system will ensure 99.5 percent availability on monthly basis.
- Maximum downtime: 4 hours/month under planned maintenance in the 2-4 AM.
- Graceful degradation to make sure that critical aspects are still available in case of non-critical service failures.

Reliability:

- ACID compliance of database including foreign key constraints implemented. Failed API calls are automatically retried (3 times with exponentially less delay).
- On payment failures, rollback of its transaction to ensure consistency of data. The critical tables should be integrity checked daily.

Maintainability:

- Separation of concerns Modular architecture (MVC pattern).
- Critical business logic coverage of over 70 percent.

- Simple updates and fixes conducted within 48 hours of detection of the problem.
- Error logs and critical failure automatic alerts.

Portability:

- Supported by the latest browsers (Chrome, Firefox, Safari, and Edge).
- Adaptive design to mobile phones, tablets and desktops.
- Minimum hardware specification: 2GB of RAM and 50mb of free storage.

Language Support:

- Toggle support of English and Bengali languages.
- E-mails (SMS, in-applications) in Bengali languages.
- Money in the format of BDT (Bengali Taka).

Backup and Recovery:

- Incremental back up every morning at 2:00 AM and full back up on weekly basis.
- Retention of 90 days with Google drive integration and verification of integrity by use of SHA-256.
- Recovery Time Objective (RTO): 4 hours; Recovery Point Objective (RPO): 24 hours.

4.3 Use Case Diagram

4.3.1 Seller (Small Business Owner)

Description: The primary user of the shop (both physical and digital). Use Cases:

- Register/Login: Register a business account and prove it with the help of OTP.
- Inventory Control: Add products, modify stocks, and price.
- Add Offline Loan: Add a loan to the walk-in customer manually (Calculator and Schedule generation).
- Accept/ reject EMI Requests: Grant and reject online EMI requests that are received in the market.
- Check Buyer KYC: Authenticate the uploaded NID/Passport photos of a buyer.
- Record Offline Payment: Recording cash or cheque receipts made in the shop.
- View Reports: View the dashboard to get access to overdue lists, daily collection, and risk score.

4.3.2 Buyer (Customer)

Description: The final consumer who is buying the merchandise. Use Cases:

- Browse Marketplace: Find products and see their information/seller. Upload KYC: Upload NID/Passport photos to their profile to be verified.
- Request EMI Loan: Search the product, tenure (months), and apply a loan request.
- Make Online Payments: Pay in installments using Stripe/bKash/Nagad/SSLCommerz.
- Check Schedule: Visit my schedule to see the next due action, the amount left to pay, and payment history.
- Download Receipt: You are offered the PDF receipt of paid installment.

4.3.3 Admin (Super-user)

Description: The platform monitor who is in charge of security and compliance. Use Cases:

- Confirm Seller Registration: Examine business documents and accept pending seller accounts.
- Dispute Management: Interferes when a Buyer complains about a Seller (or the other way round).
- Configuration of the System: certain global parameters (e.g., maximum accessible interest rate, SMS gateway parameters).

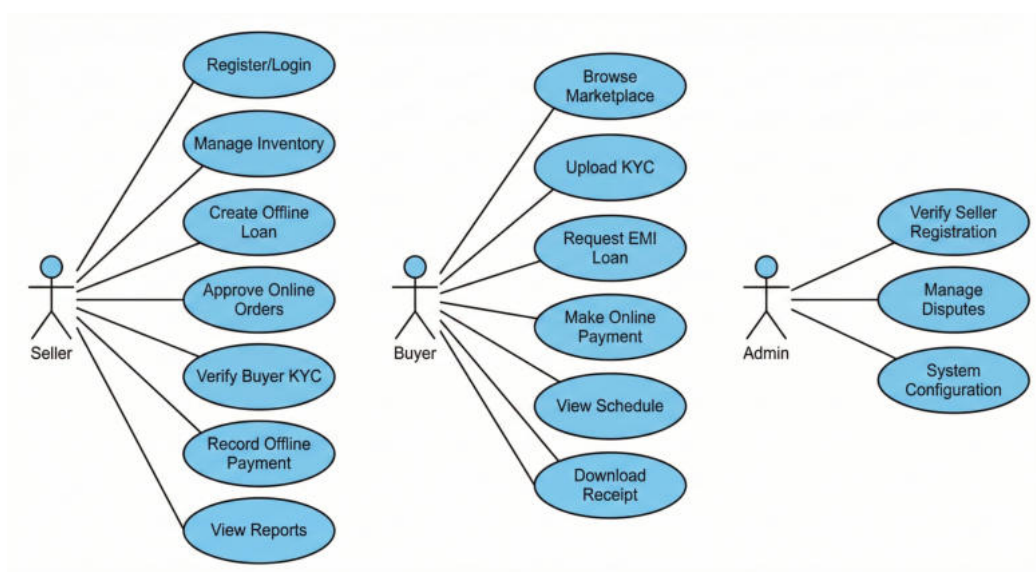


Figure 4.1: Use case diagram

4.4 Activity Diagram

The EMI Management and Loan Tracking System activity diagram displays the working process and decision-making process in the system, both in the offline sales in the shop and online in the marketplace. It traces the user experience since registration and uploading of KYC documents to loan application, approval of the seller and eventual disbursement. The diagram visualizes such key behavior of the system as automated EMI calculation, risk score assessment, payment verification, and generation of due date alerts. This assists in comprehending the rational progression of information and management among the Buyer, the Seller and the automated System Timer.

4.4.1 Activity Diagram: Offline Sale (The Replacement of the "Notebook")

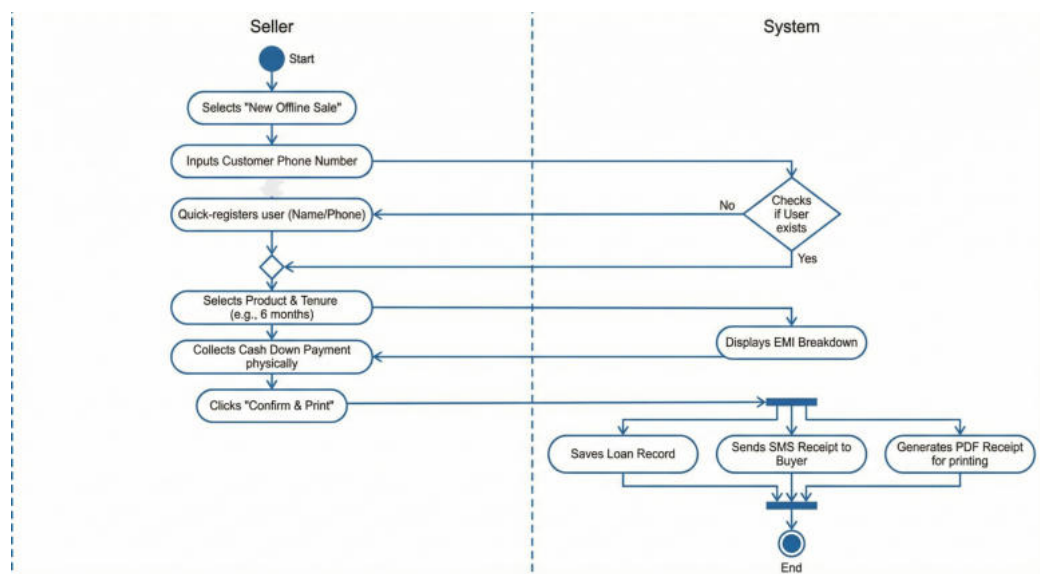


Figure 4.2: Offline sell/loan activity diagram

4.4.2 Online Loan Application (Marketplace Flow) Activity Diagram

This diagram shows the process of Buyer picking a product to the activation of the Loan.

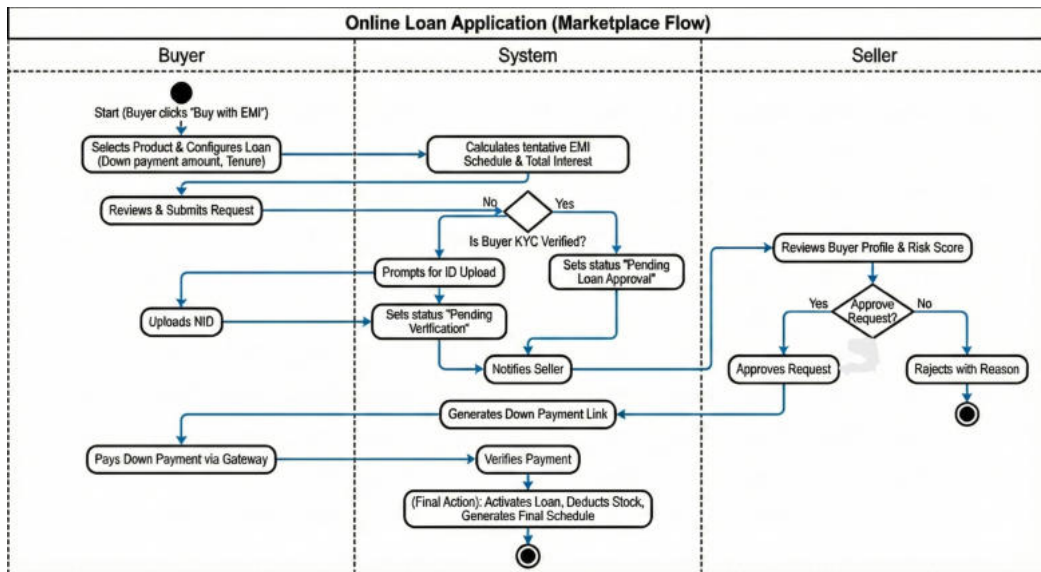


Figure 4.3: Online buy/loan activity diagram

4.4.3 Activity diagram Daily Overdue Check (The Automation)

This diagram shows the reasoning of the Cron Job (System Timer) which runs every morning. This includes the requirement of Risk Analysis and Reminder.

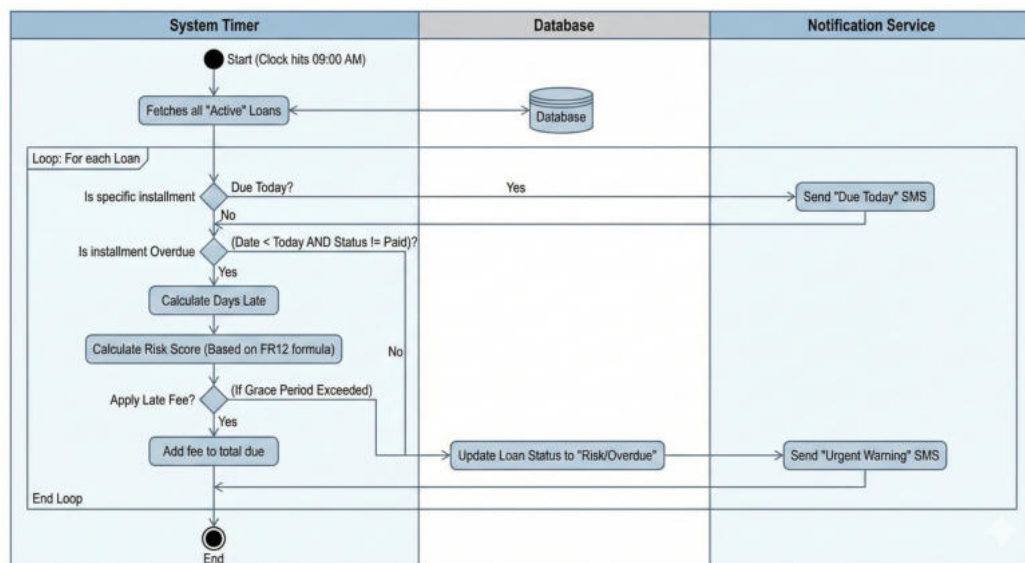


Figure 4.4: Activity diagram of daily overdue check

4.5 Data Flow Diagram

DFD is a mapping of information flow in a system. The DFD indicates the source of the data, the destination of data and the storage of data.

4.5.1 DFD Level 0: Context Diagram

This is the "Bird's Eye View." It represents the whole system as a single circle (Black Box) which is in contact with the external world.

- Central Process: EMI Management System.
- Internal Entities: User, Customer, Merchant, SMS-service.

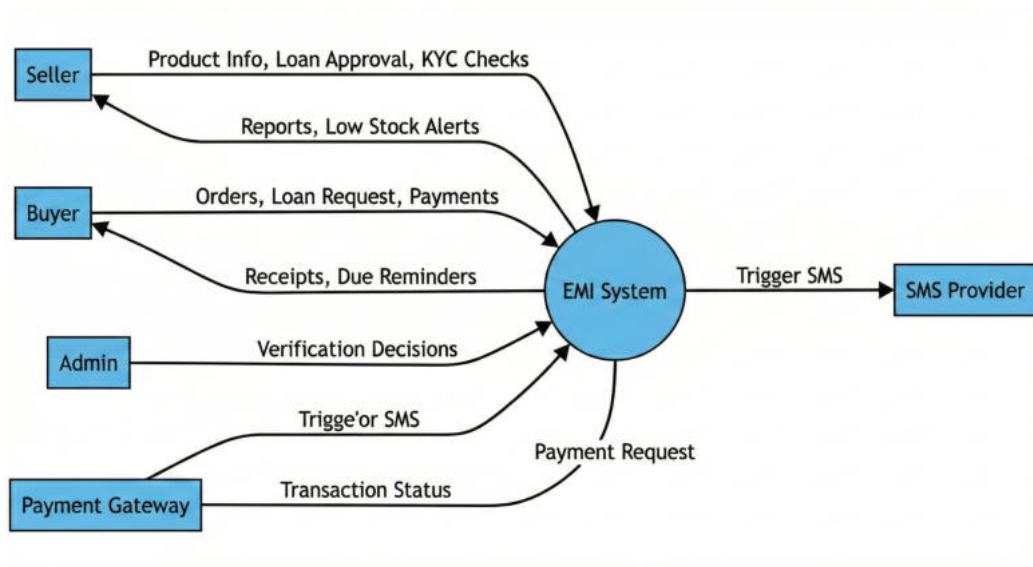


Figure 4.5: Level 0 Data Flow Diagram

4.5.2 DFD Level 1: Detailed Process Diagram

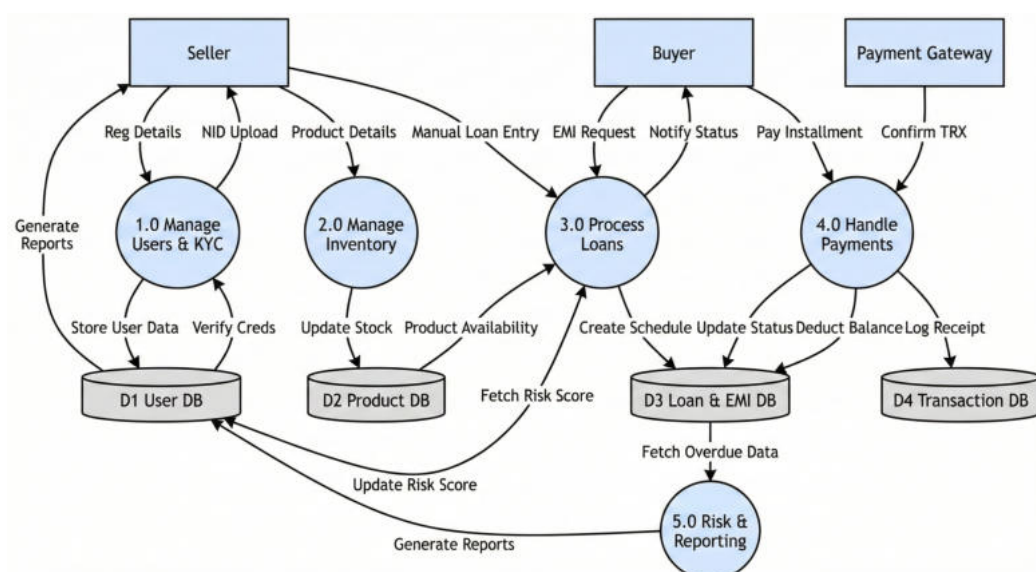


Figure 4.6: Level 1 Data Flow Diagram

Chapter 5

Design

5.1 System Architecture

5.1.1 Architectural Overview

The EMI Management and Loan Tracking System is developed on a Three-Tier Architecture (also referred to as and Client-Server Architecture). Such design provides a clear separation of concerns where the system is divided into three logical layers, the Presentation Layer (Frontend), the Application Logic Layer (Backend), and the Data Persistence Layer (Database).

The system is based on the model of communication between the client and the server using the RESTful API (Representational State Transfer) which provides the system with scaling and simplicity of integration with third-party services. Figure 5.1 shows the visualization of the architectural flow.

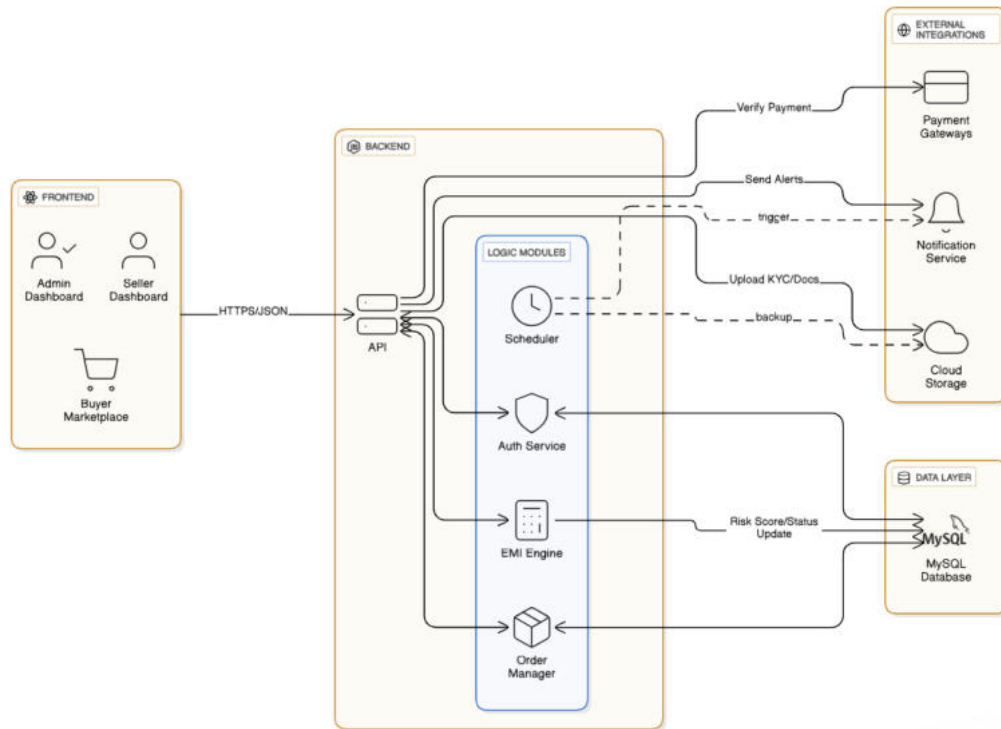


Figure 5.1: System Architecture

5.1.2 Component Description

The system is made up of four major components as discussed below:

1. The Presentation Layer (Frontend) This layer is used to deal with user interaction and interface rendering. It is loosely coupled with the backend logic, and it is executed autonomously in the browser of the user.

- **Dashboards:** Specific interfaces of Admins (system management) and Sellers (inventory and loan management).
- **Buyer Marketplace:** This is a portal that is exposed to customers where they are able to shop products and request EMI financing.
- **Technology:** React.js has been used to ensure that it has a dynamic user experience that is responsive. It exchanges the information with the backend only through `HTTPS/JSON`.

2. The Application Logic Layer The layer is referred to as the brain of the system where there is processing of business rules, calculations and security. It is coded on the basis of Node.js, and contains a few major in-house modules:

- **API Interface:** The interface that takes the input of the front end in the form of a JSON query and directs the request to the service.
- **Auth Service:** Provides user authentication (Login/Registration) and authorization services, so that only proven Sellers or Admins will be able to access sensitive data.
- **EMI Engine:** This is the computer or computing engine that does all the calculations involving the interest rates, amortization schedule, and the balances of the loan.
- **Order Manager:** Logic concerning inventory, it ensures that an order decrement when sold and is in charge of the lifecycle of an order.
- **Scheduler (Cron Jobs):** The Scheduler is an automated service that runs in the background, to perform tasks of an automated nature based on time, such as database backups and dispatching of daily alerts about payment due without the need of human intervention.

3. Database Data Persistence Layer (Database). The system should be strict on data consistency of financial records.

- **Database:** Structured information, which is User Profiles, Loans, EMI Schedules and Transaction History, is stored in a database, which is MySQL relational database.
- **Rationality:** MySQL is also ACID (Atomicity, Consistency, Isolation, Durability) compatible which is vital in averting financial errors and the maintenance of complicated ties among sellers and loans at the right degree.

4. External Integrations The system has extensions that can be integrated with third-party services the company can trust, without reinventing the wheel:

- **Payment Gateways:** The online payment can be verified in time and is integrated with bKash, Nagad, and SSLCommerz.
- **Notification Service:** This is an SMS/Email gateway provider which is attached to send auto-reminders and OTPS.
- **Cloud storage:** To avoid bloating the database, massive files such as KYC images and product photos are uploaded into a secure cloud storage system like AWS S3 or Google Drive and would only store the file reference URLs in the MySQL database.

5.1.3 Data Flow and Communication

The system has the following protocols where data passes through the system:

- **1. JSON over HTTPS:** The Frontend uses input fields to send user requests (e.g., "Create Loan") in the form of JSON objects to the Backend API.

- 2. SQL Queries: SQL Queries The Backend processes these requests and sends optimized SQL Queries to read or write to the MySQL Database.
- 3. API Webhooks: When using external services (such as Payment Gateways), the system will use webhooks to get real-time notification of the status of the transaction.

5.1.4 Design Justification

It is because the project needed a Modular Monolithic structure (Node.js + MySQL) that was decided based on the particular requirements of the project:

- 1. Cost-Efficiency: It enables the application to operate in a single server instance, which makes the hosting cost low when embracing the application by small business.
- 2. Reliability: The isolation of the Scheduler will make sure that automated tasks (reminders) are not sluggish to the user interface.
- 3. Security: The system does not allow the malicious access of sensitive financial information by direct malicious access to the Database as the Frontend is segregated of the Database through the API layer.

5.2 Database Design and Schema

5.2.1 Overview

The project database structure will be such that it effectively handles the user authentication, loan processing and financial transactions. Authentication credentials and role definitions are stored in the Users table and used as the central store of such data associated to particular SellerProfiles and BuyerProfiles. The Loans table is used to manage loans and it serves as an intermediary between buyers and sellers and products in determining the amount of principal and interest rates. The EMISchedules table gives a breakdown of monthly installment payment deadlines and Schedule of Transactions table captures the real history of the payments made to ensure transparency in financial transactions. Lastly, the KYC-Documents table is where the users are connected with their identity checks verifications, which is secure onboarding and fraud protection. This relational structure guarantees the data integrity, auditability and hybrid offline-online operations. Figure 5.2 is the Entity-Relationship (ER) Diagram.

5.2.2 Core Entity Modules

The database has been divided into four main logical modules:

1. User Management/ Authorization. To ensure that there is a clean separation between authentication credentials and role specific data, the system employs One to One relationship strategy.

- **Users Table:** This will be the main authentication table that includes the following: login credentials (email, passwordhash, phone) and system-wide information (role, isverified). This enables one system of logging in to the system by both the Admins, Sellers, and Buyers.
- **SellerProfiles and BuyerProfiles:** These are tables containing role specific attributes connected by the userid Foreign Key. Seller Profile holds such business information as shopname and tradelicenseno. Buyer Profile contains personal information (nidnumber, address), and the risk score, which is vital to the loan facilities.
- **KYCDocuments:** There is another table where sensitive identity documents (NID/Passport) are managed, which connects them to the user. This helps in boosting security and enables one to have more than one user to upload documents.

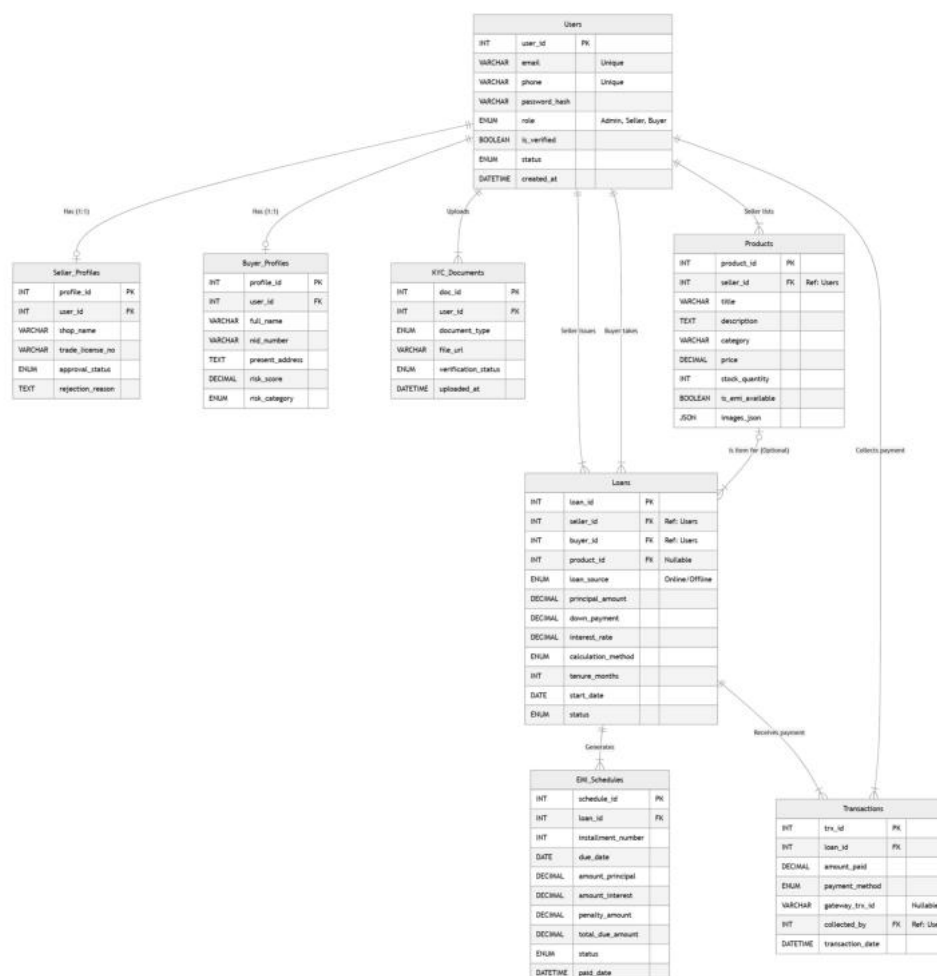


Figure 5.2: ER diagram

2. Inventory Management

- **Products Table:** It is a table where Sellers control their catalog. It has a column named `imagesjson` which is a JSON column to store several image references. It has a flag `flagemiavailable` to enable the sellers to permit EMI only on specific high-value items.

3. Loan Processing Engine Loans table is a central point in the system and it creates a connection between a Buyer, a Seller and a Product.

- **Hybrid Loan Support** The `productid` column will be Nullable. The reason behind this is a design decision to accommodate the Offline/Manual Loans whereby a seller is free to prepare a loan to a walk-in customer without having to choose a digital product that is listed.
- **Loan Attributes:** It records the snapshot of the monetary deal such as amount of principal, interest rate, and calculation method (Flat/Reducing) so that in the event of future changes in the prices of products, it will not interfere with the active loans.

4. Financial Tracking and Audit trail. The system distinguishes between Scheduled Payments and Actual Transactions so that it can be required to have strict financial accounting.

- **EMISchedules:** It is a table that reflects the anticipated cash flow. It creates rows of each installment (e.g. 6 rows in a 6-month loan) of the `duedate` and `amountdue`. The cron job of the system queries this table to create overdue alerts.
- **Transactions:** This table is used to keep account of the money received practically. It does partial payments and is connected to definite EMI schedules. It takes the `paymentmethod` (Cash/bKash) and the `gatewaytrxid` to be verified externally.

5.2.3 Key Design Decisions

- **Normalization of Profiles:** The system makes profiles and Users to be separate to enable scalability. Should a user require the capability to be a Buyer and a Seller sometime in the future, the structure allows the addition of the second profile without creating the duplicate of the log in account.
- **Auditability:** This is because the EMISchedules and Transactions are separated and therefore provide a full audit trail. The difference between What should have been paid vs. is easily calculable. What was paid to be the determinant of penalties.

- Risk Scoring Integration: With the riskcategory and riskscore directly incorporated into the BuyerProfiles table, it is now possible to make decisions based on loan approvals instantly without having to recalculate all the complicated history each time any user requests a loan.

Chapter 6

Conclusion

The implementation of the EMI Management and Loan Tracking System is a milestone in the digitalization of the small businesses in Bangladesh. The project manages to resolve the severe inefficiencies related to the manual management of loans because it eliminates paper-based notebooks and simple spreadsheets in favor of a centralized digital system.

This system is rather efficient as it achieves the main purposes of the system that is to offer a dual interface marketplace that supports the physical store sales and the online e-commerce transactions. Introduction of the automated logic i.e. interest calculation, penalty application and scoring risk drastically decreases the amount of administration on the sellers and limits the probability of human error. In addition, the secure KYC protocols and automated notifications of the Due Date add to the reliability and responsibility between the sellers and the buyers.

Although the scope will include the necessary EMI activities and inventory management, the system will be scaled. The possible improvements in the future involve the use of superior AI-based credit scoring systems and mobile application assistance in an attempt to broaden accessibility even more. Finally, the project also helps the society reach the financial inclusion objective, allowing small retailers to have the enterprise level tools to ensure their income and the business continues to grow in a competitive market.